

Module 4 — HIJING Theory

Week 4 · Soft + hard, nuclear geometry, jet quenching

4.1 Why a dedicated heavy-ion generator?

A Au+Au collision at RHIC produces ~5000 charged particles in the most central events — two orders of magnitude above minimum-bias pp. Naively running Pythia on 197 nucleons × 197 nucleons does not reproduce nuclear geometry effects (shadowing, nuclear PDFs, spectator fragments) or the collective behavior seen in data. HIJING (Heavy-Ion Jet Interaction Generator) was written by Xin-Nian Wang and Miklos Gyulassy to address these.

4.2 The HIJING model in one page

- **Nuclear geometry.** Nucleon positions sampled from a Woods-Saxon distribution. A Glauber-style calculation picks the number of participant nucleons N_{part} and binary collisions N_{coll} for each impact parameter b .
- **Soft component.** Each wounded nucleon emits “strings” whose endpoints are valence quarks and diquarks. Strings fragment via the Lund model (as in Pythia).
- **Hard component.** Each binary collision with p_T above a cutoff p_0 (~2 GeV/c) produces a mini-jet via a pQCD $2 \rightarrow 2$ matrix element, using nuclear-shadowed PDFs (EKS98-like prescription).
- **Jet quenching.** Optionally, hard partons lose energy in traversing the nuclear medium according to a parameterized dE/dx . (Switch with IHPR2(4).)
- **Spectators.** Nucleons that don’t interact form the forward/backward nuclear remnants, important for the Zero-Degree Calorimeter (ZDC) trigger.

Historical note: HIJING’s built-in jet quenching is a *phenomenological* parameterization. Modern analyses at RHIC/LHC often turn it **off** and study quenching as a separate model layer.

4.3 Centrality and impact parameter

The impact parameter b determines the geometric overlap. Experimentally, centrality classes are defined by a percentile cut on the total charged multiplicity (or on a forward-detector energy). In HIJING you request a range of b directly via the input arguments.

Typical RHIC classes:

Centrality	b range (fm)	N_{part} (Au+Au)
0-10%	0 - 4.9	~325
20-30%	6.9 - 8.3	~180
60-80%	10.7 - 13.0	~20

4.4 HIJING vs. modern alternatives

HIJING remains the workhorse for sPHENIX backgrounds, but AMPT (which takes HIJING as its input and adds hadronic rescattering), EPOS, and PYTHIA/Angantyr are all worth knowing. For this course we stick with HIJING because of its direct support in Fun4All.

4.5 Reading

- Wang & Gyulassy, Phys. Rev. D 44 (1991) 3501 — original paper.
- Gyulassy & Wang, Comp. Phys. Comm. 83 (1994) 307 — code manual.
- Miller et al., *Glauber modeling in high-energy nuclear collisions*, Ann. Rev. Nucl. Part. Sci. 57 (2007) 205.

4.6 Written exercise

Write a 1-page summary (in your own words) answering: (a) what is the mini-jet model, (b) why does centrality matter for physics observables, (c) what is nuclear shadowing and why does it modify the PDF at small x ?

4.7 Self-assessment

- ☐ I can distinguish N_{part} , N_{coll} , and impact parameter b .
- ☐ I can sketch the Glauber picture on a napkin.
- ☐ I understand why a pure Pythia-based A+A simulation is inadequate.